# ALCHEMUS

## Advanced Drug Analysis Platform

### Research Owner Wallet

26zfAn2dw6gNzb2yNVECGLBxgGxPG2qJdY3pKRTk2oX1

Date: 5/1/2025
Compounds: Omeprazole, Salbutamol, Cocaine



## New Compound Analysis

Name: Cocasultazole
Formula: $C_{32}H_{42}N_4O_6$
Weight: 582.71
Drug-likeness: 0.67

Activity:
Bronchodilator and gastric acid secretion inhibitor with local anesthetic properties

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## Target Analysis

Dopamine transporter (DAT)
Binding Affinity: 15.2
Primarily responsible for cocaine's reinforcing effects

Proton pump (H+/K+ ATPase)
Binding Affinity: 3.2
Reduction of gastric acid secretion

Beta-2 adrenergic receptor
Binding Affinity: 6.5
Bronchodilation in asthmatic patients

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## Toxicity Profile

Risk Level: High
Affected Systems: heart, lungs, stomach, brain

Mitigation Suggestions:
- monitor cardiovascular health
- prescribe antacids for gastric side effects
- ensure patient is not on sympatomimetic drugs

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## Pharmacokinetics

Absorption:
Rapidly absorbed in the gastrointestinal tract

Distribution:
High plasma protein binding

Metabolism:
Hepatic metabolism primarily via CYP2C19

Excretion:
Mainly renal excretion

Half-life: Approximately 2-8 hours

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## Detailed Analysis

Cocasultazole is a novel compound that merges the functionalities of Cocaine, Omeprazole, and Salbutamol. This compound offers a unique approach to managing combined symptomatic manifestations such as bronchoconstriction and gastric acid secretion. By inhibiting the dopamine transporter, the compound harbors potential efficacy in dopaminergic modulation but retains risks seen in cocaine analogs. The inclusion of proton pump inhibition adds the capacity to manage gastric conditions. The bronchodilator function inherited from Salbutamol offers therapeutic effects for acute asthmatic conditions. However, potential cardiovascular risks due to norepinephrine modulation highlight an urgent need for cautious clinical evaluation. Future research to develop analogs that maintain therapeutic benefits while minimizing adverse reactions is essential.

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